SUPPORTING ADULT GUIDANCE

The angle at the centre

Circle Theorems · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This is the **keystone** lesson of the unit: the theorem that the angle subtended by an arc at the centre is twice the angle it subtends at the circumference. Learners first **discover** the 2 : 1 (and the invariance of the inscribed angle) by dragging a point in the Circle Theorem Lab, then **prove** it via the isosceles-triangle argument. Every subsequent theorem in the unit — angle in a semicircle, angles in the same segment, cyclic quadrilaterals — is deduced from this one, so securing both the result and its reason matters greatly.

Driving Question: *Why is the angle at the middle always double the one at the edge?*

The mathematics, in plain terms

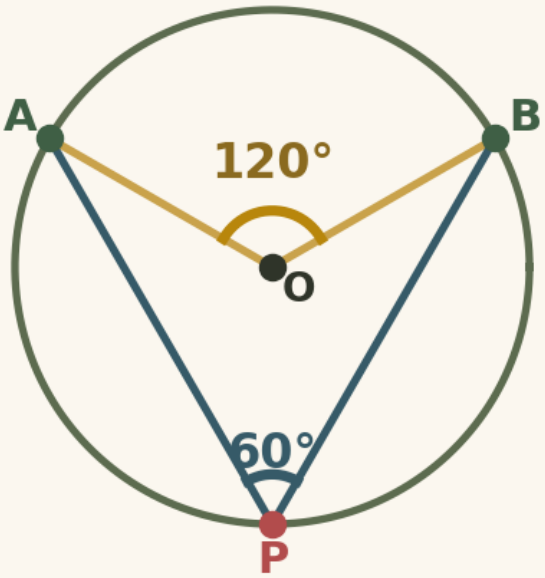
For an arc AB, the central angle ∠AOB equals twice any inscribed angle ∠APB on the major arc. Proof: OA, OB, OP are radii; triangles OAP and OBP are isosceles, so each apex angle is 180 – 2(base angle); and since P, O, Q are collinear, the angles at O on that line give ∠AOQ = 2x and ∠BOQ = 2y, so ∠AOB = 2∠APB. When P is on the minor arc the central angle is the reflex angle, and the relationship still holds. A direct corollary is that inscribed angles on the same arc are equal (angles in the same segment).

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Use the theorem both ways

O is the centre. The angle at the centre ∠AOB is 120°. Find the angle at the circumference ∠APB.

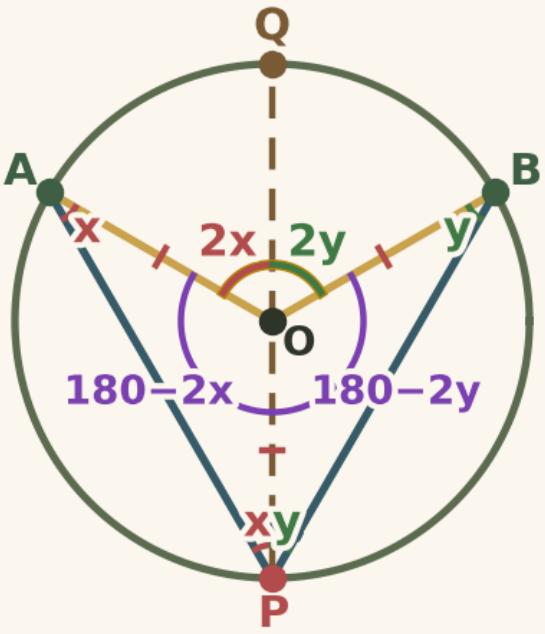


1. Both angles stand on the same arc AB.	∠AOB and ∠APB on arc AB
2. Angle at the centre = 2 × angle at the circumference.	120 = 2 × ∠APB
3. So halve the centre angle.	∠APB = 120 ÷ 2

∠APB = 60°, because the angle at the centre is twice the angle at the circumference on the same arc.

Example 2 — Why it is exactly double

Prove ∠AOB = 2 × ∠APB. Join P to O and extend it to Q.



1. OA = OB = OP are radii, so triangles OAP and OBP are isosceles.	equal radii
2. Isosceles ⇒ equal base angles: ∠OAP = ∠OPA = x and ∠OBP = ∠OPB = y.	∠APB = $x + y$
3. Angles in each triangle sum to 180°, so the apex angles are ∠AOP = 180 – 2x and ∠BOP = 180 – 2y.	apex angles
4. P, O, Q lie on a straight line, so ∠AOQ = 180 – (180 – 2x) = 2x, and ∠BOQ = 2y.	∠AOQ = 2x, ∠BOQ = 2y

∠AOB = 2x + 2y = 2(x + y) = 2 × ∠APB. The angle at the centre is twice the angle at the circumference. ■

Discover, then justify — always with a reason

Let the learner drag the point and *see* both the constant inscribed angle and the 2 : 1 before any rule is stated; conviction first. Then build the proof one line at a time, leaning on the isosceles base-angle fact (keep a Reconnection Route to it open). Insist on the reason with every numerical answer — 'angle at the centre is twice the angle at the circumference' — because reasoning, not just the number, is assessed throughout the unit. Introduce the reflex case only once the standard case is secure.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Name the centre angle and the circumference angle on the same arc.	~9 min
Movement	Halve or double to find a missing angle, with the reason.	~10 min
Reflection	Circle Theorem Lab: drag P, see the constant angle and the 2 : 1.	~13 min
Creativity	Prove it with the isosceles triangles; meet the reflex case.	~9 min
Rest	A short pause on invariance and a proof that closes.	~2 min
Nutrition	Intellectual: the power of an invariant and of a proof.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Doubling when they should halve (or vice versa).
Anchor to which angle is given: 'Is the known angle at the centre or the circumference? Centre is the big one — halve it to get the edge.'

Giving a number with no reason.
'Correct — now what is your reason? Name the theorem: the angle at the centre is twice the angle at the circumference on the same arc.'

Pairing angles that stand on different arcs.
'Do these two angles really stand on the SAME arc? Trace the arc each one opens onto before you halve or double.'

Thinking the theorem fails for a reflex centre angle.
'When P is on the minor arc, use the reflex angle at the centre — it is still exactly twice ∠APB.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the parts of a circle and 'standing on an arc' (Lesson 1), angles in a triangle sum to 180°, and isosceles base angles are equal. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner doubles a centre angle, or gives 60° with no justification. The aim is a question that restores the direction of the rule and the habit of a reason.

- "Is the angle you were given at the centre or at the circumference?"

- “Centre is twice circumference — so do you halve or double here?”
- “What is your reason? Which theorem are you using?”
- “Do both angles stand on the same arc? Show me the arc.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why is the inscribed angle the same wherever P is?

Because it always equals half the same central angle ∠AOB. Since ∠AOB is fixed by A and B, every inscribed angle on that arc is half of it — so they are all equal. This is the 'angles in the same segment' fact, met properly next lesson.

How do I remember which way round it is?

The centre angle is the bigger one (it is twice), so halving takes you from centre to circumference and doubling takes you the other way. Saying 'centre is the big one' fixes the direction.

Does it work when P is on the small arc?

Yes. Then the angle at the centre on the same arc is the reflex angle, and the reflex ∠AOB is still exactly twice ∠APB.

Why prove it if the Lab already shows it?

The Lab convinces; the proof explains and guarantees. Every later theorem in the unit is deduced from this one, so a secure proof — equal radii, isosceles base angles, exterior-angle fact — is what the rest is built on.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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